MILITARY SPECIFICATION

THERMOMETER, SELF-INDICATING, 5-INCH SCALE SIZE, LIQUID-IN-GLASS, MERCURY

1. SCOPE

1.1 Scope.- This specification covers self-indicating, 5 inch scale size, liquid-in-glass, mercury thermometers.

1.2 Classification.- Thermometers shall be of the following forms as specified (see 6.2 and figure 1).

Form A - Straight stem
Form B - 90° Back angle
Form C - 90° Left side angle
Form D - 90° Right side angle
Form E - 45° Reclined angle

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-636 - Box, Fiberboard.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive, Paper, Water-Resistant (for Carton Sealing).
PPP-T-97 - Tape, Pressure-Sensitive Adhesive, Filament-Reinforced.

MILITARY
MIL-P-77 - Plastic Sheet and Plastic Rod, Thermosetting, Cast.
MIL-P-116 - Preservation, Methods of.
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use.
MIL-P-5425 - Plastic, Sheet, Acrylic, Heat-Resistant.
MIL-L-10547 - Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.
MIL-E-15090 - Enamel, Equipment, Light-Grey, (Formula No. 11).

STANDARDS

MILITARY
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-735 - Test Methods and Test Equipment for Thermometers Used in Machinery and Piping Systems.

HANDBOOK

MILITARY
MIL-HDBK-H-1 - Cross-Index of Chemically Equivalent Specifications and Identification Code (Ferrous and Non-ferrous Alloys).

(See Supplement-1 for list of applicable MS Standards).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6685

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A-276-62 - Hot Rolled and Cold Finish Corrosion-Resisting Steel Bars.
B-21-62 - Naval Brass Rods, Bars and Shapes.
B-36-61 - Brass Plate, Sheet, Strip, and Rolled Bar.
B-62-60 - Composition Bronze or Ounce Metal Castings.
B-85-60 - Aluminum-Base Alloy Die Castings.
B-209-62 - Aluminum-Alloy Sheet and Plate.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Service.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, New York.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials.- Thermometer parts shall be in accordance with table I. Wells shall be as specified (see 6.2). Chemically equivalent metal materials in accordance with MIL-Handbook H-1 may be substituted.

Table I - Materials.

Thermometer part	Material	Material specification	Remarks
Case and frame	Brass	ASTM-B-62	Case and frame shall be painted grey in accordance with MIL-E-15090 (formula No. 11)
	Mallable iron	As approved	
	Cast steel	As approved	
	Cast aluminum	ASTM-B-85	
Scale	Cres	ASTM-A-276	Scale finish shall be a white enamel or silvered background with black enamel graduation, baked on for 30 minutes at 300°F. or equivalent. Scale shall be held in place by 2 screws and shall be adjustable over not less than 2 minor graduations.
	Aluminum	ASTM-B-209	
	Brass	ASTM-B-36	

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Table I - Materials (Cont'd.).

Thermometer part	Material	Material specification	Remarks
Tube	Glass	–	See 3.2.1
Bulb	Glass	–	Corning normal glass or equivalent
Window	Plastic	MIL-P-77(G-3), MIL-P-5425 or HCR-39	Shall be at least 1/16 inch thick
Indicating medium	Mercury	–	–
Protective tube	CRES	ASTM-A-276	Protective tube shall be of a one-piece construction
Jam nut	Brass	ASTM-B-21	See figure 2

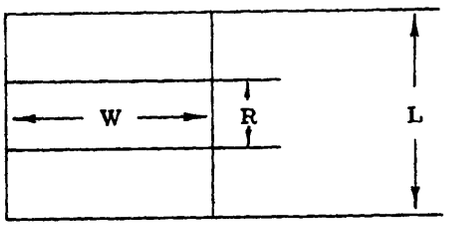
3.1.1 Heat transfer medium.- A heat transfer medium may be used to surround the mercury filled bulb. The medium shall not polymerize or change its state at 50°F above or below the dial range. The medium shall have no adverse effects on the parts to which it is exposed.

3.2 Design.- Thermometers shall be furnished in the forms shown on figure 1 and shall consist of a case and frame, scale, glass tube and bulb, indicating medium, window, jam nut (see figure 2) and protective tube (see figures 3 and 4).

3.2.1 Glass tube and bulb.- Visible portion of the glass tube shall be red reading and magnifying lens type, with the smallest dimension of the cross section not less than 4 millimeter (mm). It shall be filled with inert gas to aid in the prevention of mercury separation and have an expansion chamber blown at the top of the tube to prevent breakage in the event of overtemperatures up to 50°F. The tube and bulb shall be sufficiently aged to prevent volumetric changes after being placed in service.

3.2.2 Scale.- Scale dimensions shall be in accordance with table II.

Table II - Scale markings.



Nominal length (L) 1/	Range (R) 2/ (minimum)	Width (W) 3/ (minimum)
Inches	Inches	Inches
5	3-1/2	1

1/ L - Length of scale.
2/ R - Scale range as specified in table III.
3/ W - Unobstructed window width.

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3.2.2.1 Markings and graduations.- The scale shall be graduated in accordance with table III. The units (°F), manufacturer's name or trademark and the legend mercury shall be legibly marked on the scale.

Table III - Scale graduations.

Range of scale °F	Accuracy °F	Number of degrees between graduations	Number of degrees between figures
-40 to 110	1.5	2	20
30 to 240	2.1	2	20
50 to 400	3.5	5	50

3.2.3 Threads.- Threads shall be in accordance with H28. Taper pipe threads are not acceptable.

3.3 Performance.- Performance shall be in accordance with table IV.

Table IV - Performance requirements.

Test	Requirement	Test procedure
Accuracy	± 1 percent of the range span (see table III)	4.7.1
Load	100 pounds	4.7.3
Thermal response (lag)	Shall not exceed 12 seconds lag	4.7.2
Vibration	± 1 percent of the range span	4.7.4
Thermal and mechanical stability	Drift shall not exceed 1/4 percent of the range span after 96 hours. However if the drift is greater than 1/4 percent but less than 1 percent of the range span, the test shall be extended 288 hours to a total of 394 hours and then terminated. If the thermometer shows a drift of less than 1 percent of the range span it shall be acceptable.	4.7.5

3.4 Drawings.- Master drawings shall be in accordance with class A of MIL-D-963 and shall cover all designs in accordance with 3.2. Drawings shall include the following:

(a) Sectional views and details necessary to verify conformance to this specification.
(b) Materials of parts.
(c) Dimensions required to assure interchangeability.
(d) Operating data, ranges, scale markings and other data required for proper selection, for example pressure rating of wells.
(e) Test approval data, including approval letters, report numbers and dates.
(f) Complete weight.
(g) Indicate special consideration which may effect service selection.
(h) Performance data.
(i) Identify quality control document showing conformance with the appendix of this specification and reference the acceptance letter.
(j) Critical quality control procedures (see 4.1.2).
(k) Quality control symbols (see 4.1.3).

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3.4.1 Copies of drawings.- Unless otherwise specified, two copies of the drawing shall be submitted for initial approval to the Bureau of Ships. For subsequent procurement, the number of copies shall be specified in the contract or order.

3.4.2 Approval of drawings.- Drawings shall be approved by the Bureau of Ships. Approval shall be required only on the first submittal of an item and whenever changes are made. Where drawing approval has previously been obtained, it shall be necessary only to submit copies of the approved drawing to meet drawing requirements on subsequent contracts.

3.5 Workmanship.- Contractor's production and inspection processes shall be established so that the end results will produce a completed product in strict compliance with this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspection set forth in this specification; where such inspections are deemed necessary to assure supplies and services conform to perscribed requirements.

4.1.1 Quality control system.- The contractor shall provide and maintain a quality control system acceptable to the Government. The system of quality control shall be in accordance with the appendix of this specification and as specified in 4.1.2. The contractor is in no way relieved of the final responsibility of furnishing thermometers meeting the requirements of this specification.

4.1.2 Critical quality control procedures.- Critical procedures shall be written and maintained up-to-date for use in training of and by production personnel. Procedures shall be in accordance with 30.2 of the appendix to this specification.

4.1.3 Quality control symbols.- A quality control symbol shall be assigned to each thermometer part listed on the master drawing. The symbols shall have the following legend:

Symbol	Legend
A	Actual test reports are available. Test data may include material physicals and chemicals and when applicable performance test data.
B	Certification of compliance may be used to aid in determining whether parts meet specification requirements provided that certificate contains: (a) Objective test and examination data generated by the supplier or (b) Where data in (a) is not readily available or cannot be provided economically, as that from a controlled process, the location of the data and method of obtaining it shall be included. Certification of compliance which contains no actual test, examination of other verifiable quality evidence is not acceptable.
C	Test reports on certification of compliance not required for parts. Visual examination for materials and workmanship is satisfactory.

4.2 Pre-award survey.- When a manufacturer is to be considered for an award of a contract, the capabilities of his production facilities and evidence relative to vibration resistance of his design shall be furnished, prior to award, to assure capability of meeting all contract requirements including delivery dates.

4.3 Master drawing approval.- Master drawing approval shall consist of the following steps:

I - Approval of manufacturer's facilities in accordance with MIL-STD-735 by the Naval Boiler and Turbine Laboratory (NBTL).
II - Evaluation of quality control system to determine compliance with the appendix to this specification. This evaluation shall be accomplished by the local Government inspector.
III - Successful test reports from NBTL on sample thermometers to determine compliance with the requirements of table IV.
IV - Drawings shall be submitted for approval of the Bureau of Ships through the local Government inspector.

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4.3.1 Certification.- Steps I through IV of 4.3 shall be accomplished concurrently. A copy of the quality control system evaluation letter shall be furnished to the Bureau of Ships.

4.3.2 Tests.- Master drawing approval tests shall include all tests listed in 4.7 and shall meet the requirements of table IV.

4.3.2.1 Samples for master drawing tests.- Two sample thermometers representing form B, form E, and forms A, C, or D shall be submitted. The sample shall include the highest and lowest temperature range for which approval is desired. All samples shall have been tested for calibration and load prior to delivery to laboratory.

4.4 Quality conformance inspection.-

4.4.1 Lot (parts, subassemblies and wells).- For purposes of inspection, a lot shall consist of all parts, subassemblies and wells offered for delivery at one time under the same contract or order. The wells shall be grouped together according to material and insertion length.

4.4.2 Lot (thermometers).- For purposes of inspection, a lot shall consist of all thermometers offered for delivery at one time, grouped as follows:

Form B
Form E
Form A, C or D

4.4.3 Sampling for examination, materials and interchangeability.- A random sample of similar parts, subassemblies and wells shall be selected from each lot (see 4.4.1) in accordance with table V for the examination specified in 4.5.1. (The acceptance quality level (AQL) equals 1-1/2 percent defective.)

Table V - Sampling for examination, materials and interchangeability.

Number of similar parts, subassemblies and wells in a lot	Number of similar parts, subassemblies and wells in sample	Defectives	
		Acceptance number	Rejection number
Up to 10	All	--	--
11 to 25	10	0	1
26 to 40	15	0	1
41 to 65	25	1	2
66 to 110	35	1	2
111 to 180	50	2	3
181 to 300	75	3	4
301 to 500	110	4	5
501 to 800	150	5	6
801 to 1300	225	8	9
1301 to 3200	300	10	11
3201 to 8000	450	14	15
8001 to 15,000	750	20	21

4.4.4 Sampling for assembled thermometer inspection.- A random sample of assembled thermometers shall be selected from each lot (see 4.4.2) in accordance with table VI for the inspection specified in 4.5.2. The AQL equals 4 percent and provides for an average outgoing quality limit (AOQL) of 5 percent.

4.4.5 Sampling for verification of production inspection.- One to three, out of every 100 thermometers shall be selected at random from the production line and subjected to the inspection specified in 4.5.3.

4.5 Inspection.-

4.5.1 Examination of parts.- Each sample selected in accordance with 4.4.3 (see table V) shall be examined to determine conformance to the requirements of this specification and the master drawing. Any unit containing one or more defects shall be rejected and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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Table VI - Sampling for assembled thermometer inspection.

Number of assembled thermometers (cumulative)	Number of assembled thermometers (increments)	Sampling rate	Sample cumulative
Up to 7	7	1/4	2
17	10	1/5	4
35	18	1/6	7
56	21	1/7	10
106	50	1/10	15
181	75	1/15	20
356	175	1/25	27
756	400	1/50	35
1,556	800	1/80	45
3,556	2000	1/100	65
7,156	3600	1/120	95
14,656	7500	1/150	145

4.5.2 Inspection of assembled thermometers.- Each sample selected in accordance with 4.4.4 (see table VI) shall be inspected to determine conformance to the requirements of the specification and the classification of defects in table VII. When one sample fails the inspection at other than the highest sampling rate, sampling shall revert to the first line of table VI. However when the cumulative sample exceeds 35, one failure shall be allowed and when the cumulative sample exceeds 65, two failures shall be allowed before reverting to the first line of table VI. If one sample fails during sampling at the highest rate (1/4) the Bureau of Ships shall be notified for reconsideration of drawing approval.

Table VII - Classification of defects.

Categories	Defects
Critical:	
1	Glass tube and bulb improperly heat treated or aged during production.
Major:	
101	Evidence of unauthorized materials used.
102	Thermometer parts distort under load test.
103	Design not in accordance with the master drawing.
104	Glass tube, non-magnifying, non-red reading type, reading angles less than 10 degrees, expansion chamber for at least 50°F overtemperature not included.
105	Mercury column not continuous and unbroken.
106	Wells and bulbs noninterchangeable.
107	Packaging requirements not conforming to the specification
108	Unobstructed window width less than 1 inch.
109	Nominal scale length not 5 inches.
110	Scale range less than 3-1/2 inches.
Minor:	
201	Finish nonconforming, scratched, chipped, burrs not removed, wrong color.
202	Dial markings not included, wrong color background.
203	Scales not thick enough to prevent warping.

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4.5.3 Verification of production inspection.- Samples selected in accordance with 4.4.5 shall be thoroughly inspected to determine conformance to the requirements of the master drawings and specification and the classification of defects. When only one thermometer is selected and it fails, two others shall be selected and inspected. If two failures occur during inspection, production shall be terminated until corrective action has been taken. Successful thermometers may be delivered.

4.5.4 Deficiency reports.- Whenever required as a result of a deficiency reports from using activities, sample thermometers from production or from the delivered product shall be subjected to any inspection necessary to determine the reason for failures. The inspection shall be conducted at the Naval Boiler and Turbine Laboratory.

4.5.5 Corrective action.- Deficiencies that are not corrected within 90 days of the failure date and approved by the Bureau of Ships and the local Government inspector shall be reason for cancelling the master drawing approval. Equipment shall not be resubmitted for approval until substantial evidence that the product is in accordance with specification requirements is furnished. Use shall be made of feedback data generated and furnished by using activities as well as that generated in the contractor's facility.

4.6 Quality conformance tests.-

4.6.1 Assembled thermometer tests.- Each sample selected in accordance with 4.4.4 shall be subjected to the calibration test specified in 4.7.1.

4.6.2 Verification of production tests.- Samples selected in accordance with 4.4.5 shall be subjected to the following test sequence:

(a) Calibration (see 4.7.1).
(b) Vibration (see 4.7.4).
(c) Accuracy verification (see 4.7.1.1).

4.7 Test procedures.- Tests shall be conducted in accordance with MIL-STD-735, this specification and without the well, except where specified.

4.7.1 Calibration test.- Calibration tests shall be conducted as specified in MIL-STD-735 and shall include the following:

(a) Maximum dial indication shall conform to table III.
(b) Insertion length shall conform to figures 3 and 4.
(c) Accuracy shall conform to table III.
(d) Master standard accuracy shall not exceed ± 1/4 percent of the range span.

4.7.1.1 Accuracy verification.- This shall be the same as the calibration test except that only three points on the dial need be checked. One point shall be in the lower 25 percent of the range span, one at 50 percent and one in the upper 25 percent. This test shall be used, except for the initial calibration test.

4.7.2 Thermal response (lag) test.- Thermal response shall be determined in accordance with MIL-STD-735. Test temperature and bath medium shall be in accordance with table VIII. The response time shall be in accordance with table IV.

Table VIII - Test temperature and bath mediums.

Dial range	Bath liquid	Initial temperature	[1/] T_1 start timing	[2/] T_2 stop timing	Bath temperature
°F.		°F.	°F.	°F.	°F.
-40 to 110	Water	35	50	78	95
30 to 240	Water	50	100	163	200
50 to 400	Water	50	100	163	200

[1/] Plus or minus 5°F.
[2/] Maximum value.

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4.7.3 Load test.- The thermometer, while mounted into it's well shall withstand the load requirements of table IV applied in any direction at right angles to the case, without damage to any part of the thermometer.

4.7.4 Vibration test.- The vibration test shall be in accordance with MIL-STD-735. As a result of the test no part of the thermometer shall become loosened, dislodged or broken and parts shall show no signs of wear. A calibration check shall meet the requirements of table IV.

4.7.5 Thermal and mechanical stability.- Thermal and mechanical stability test shall be conducted as specified in MIL-STD-735 and shall include the following:

(a) Drift.- The thermometer shall be tested for drift in accordance with MIL-STD-735 and shall meet the requirements of table IV.

5. PREPARATION FOR DELIVERY

5.1 Packaging.-

5.1.1 Level A.- Thermometers and thermometer wells shall be individually packaged and cushioned in accordance with method III of MIL-P-116.

5.1.2 Level C.- Packaging shall be sufficient to affort adequate protection against physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice.

5.2 Packing.-

5.2.1 Level A.- Thermometers and thermometer wells packaged as specified shall be packed in overseas type, wood-cleated fiberboard, corrugated or solid fiberboard or wood-cleated veneer paper overlaid boxes conforming to PPP-B-591, PPP-B-636 (class 2) or PPP-B-576 respectively, at the option of the supplier. Shipping containers shall have case-liners conforming to MIL-L-10547 and shall be closed and sealed in accordance with the appendix thereto. Case-liners for boxes conforming to PPP-B-636 may be omitted, provided all joints and corners of the boxes are sealed with 1-1/2 inch wide tape conforming to PPP-T-76. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto, except that fiber boxes shall be banded with tape conforming to type III of PPP-T-97 and appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2 Level C.- Thermometers and thermometer wells packaged as specified shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or other regulations as applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order on herein, interior and exterior shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The thermometers specified herein are intended for applications within the limit of range and accuracy (see table III).

6.2 Ordering data.- Procurement documents should specify the following:

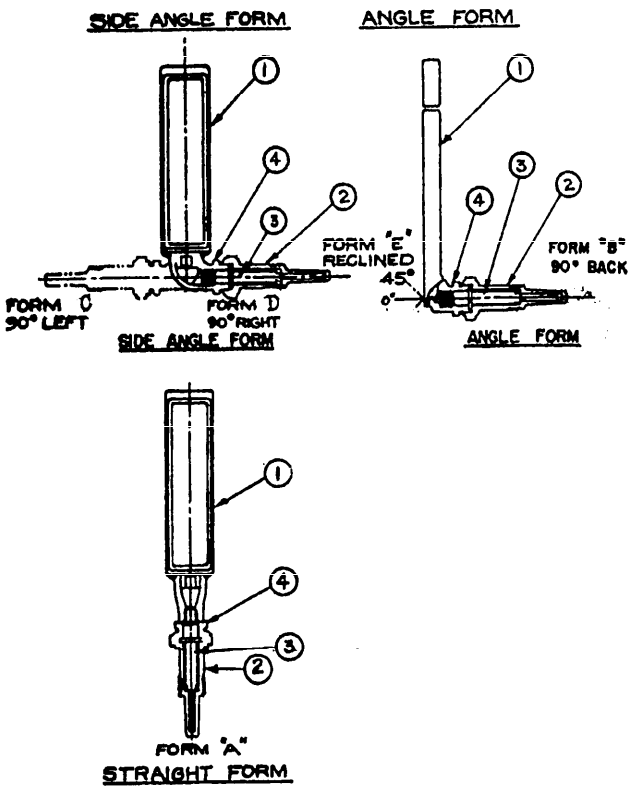
(a) Title, number and date of this specification.
(b) Form required (see 1.2).
(c) When wells are required (see 3.1).
(d) Range of scale graduations (see table III).
(e) Design of well (see Supplement for Applicable MS Standards).
(f) Insertion length (see figures 3 and 4).
(g) Copies of drawings required (see 3.4.1).
(h) Level of packaging and packing required (see 5.1 and 5.2).
(i) Requirements in the appendix (see 6.3).

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6.3 Appendix.- The appendix is to be used as a part of contractual documents by reference in the contract as a basis for the establishment of workable procedures within the contractor's facility and between the contractor and sub-contractors for the control of quality supplies or services. When made a part of sub-contracts, all provisions of this specification which apply to the contractors, shall apply equally to the sub-contractors (see 6.2(i)).

Preparing activity:
Navy - SH
(Project 6685-N072Sh)

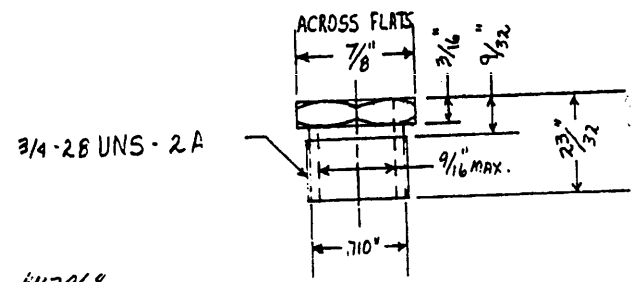
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Piece No.	Name
1	Case
2	Well
3	Bulb Chamber
4	Jam Nut

Figure 1 - Forms of thermometers and wells (forms A through E).



SH7968

Figure 2 - Jam nut dimensions.

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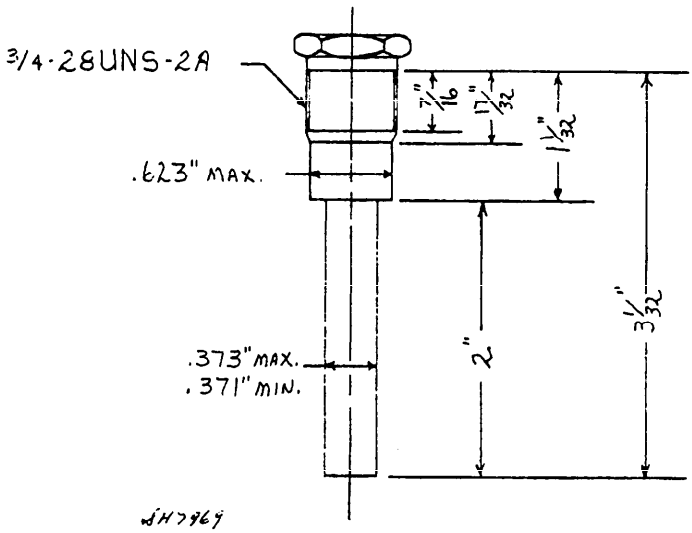


Figure 3 - Protective tube and bulb dimensions, 2-inch insertion length (without well).

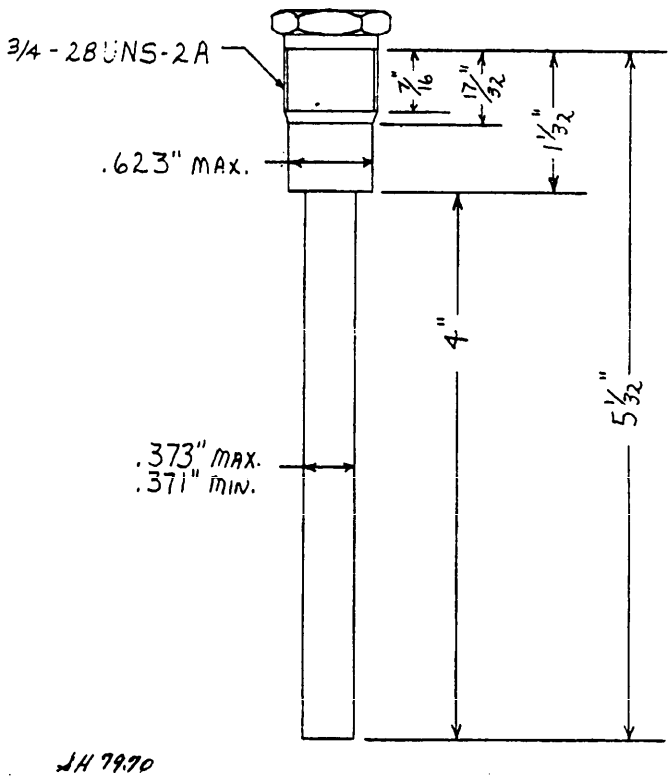


Figure 4 - Protective tube and bulb dimensions, 4-inch insertion length (without well).

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APPENDIX

QUALITY CONTROL SYSTEM

10. SCOPE

10.1 Scope.- This appendix requires the establishment of a quality control system by the contractor to assure that supplies or services meet the quality standards established by this specification. This system including procedures, is subject to surveillance by the Government representative. The procedures shall be designed by the contractor. The contractor's procedures used to implement the requirements of this appendix shall be subject to the disapproval of the Government representative.

10.1.1 Relation to MIL-I-45208 and MIL-C-45662.- This appendix contains requirements in excess of those in MIL-I-45208 and MIL-C-45662, inasmuch as total conformance to contract and specification requirements is obtained best by controlling work operation, manufacturing processes as well as examination and tests.

20. APPLICABLE DOCUMENTS

20.1 The following documents of the issue in effect on date of the invitation for bids or request for proposal form a part of this appendix to the extent specified herein.

SPECIFICATIONS

MILITARY
MIL-I-45208 - Inspection System Requirements.
MIL-C-45662 - Calibration System Requirements.

STANDARDS

MILITARY
MIL-STD-109 - Quality Assurance Terms and Definitions.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. REQUIREMENTS

30.1 Outline.- The contractor shall maintain an effective and economical quality control system planned and developed in conjunction with other planning functions. The system shall be based upon consideration of the quality under procurement, interchangeability and reliability requirements, and manufacturing techniques. The system shall assure that adequate control of quality is maintained throughout all areas of contract performance, including as applicable, the receipt, identification, stocking and issue of material, and the entire process of manufacture, packaging, shipping, storage and maintenance. All supplies or services under the contract, whether manufactured or performed within the contractor's plant or at any other source, shall be subject to control at such points as necessary to assure conformance to contractual requirements. The system shall provide for the prevention and ready detection of discrepancies and for timely and positive corrective action. The contractor shall make objective evidence of quality conformance as defined in MIL-STD-109, readily available to the Government representative.

30.2 Description of procedures.- The contractor shall provide and maintain a description of procedures for control of quality. To the extent necessary, written examination and test procedures shall be prepared to supplement the applicable drawings and specifications, and shall make clear the manner in which such examination and test procedures are to be used. The description may be a compilation of existing shop travelers, routing cards, inspection method sheets, test procedures, route sheets, or other documents normally used by the contractor to define inspection operations. The description of the quality control system and all applicable examination and test procedures shall be available to the Government representative.

30.3 Drawing and change control.- A procedure shall be maintained by the contractor to assure that the latest applicable drawing technical requirement and contract change information will be available at the time

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and place of contractor inspection. Concurrently with the effectivity of revised drawings or changes, the contractor's drawing and change control shall assure that obsolete information is removed from all points of issue and use. All changes shall be processed in a manner which will assure accomplishment on the affected supplies at the specified effective points. The contractor shall maintain a record of the point of effectivity of changes. This record shall be available for ready reference by the Government representative.

30.4 Measuring and testing equipment.- Unless otherwise specified in the contract, the contractor shall provide and maintain gages and other measuring and testing devices necessary to assure that supplies conform to contract requirements. These devices shall be calibrated against measurement standards or designated measuring equipment at established periods to assure continued accuracy. The contractor shall prepare and maintain a written schedule for the maintenance and calibration of such equipment based on type, purpose and degree of usage.

30.4.1 Production tooling used as a media of inspection.- When production jigs, fixtures, tool masters and other such devices are used as media of inspection, they shall be initially inspected or, by other suitable means, proved for accuracy prior to release for production use. These devices shall be reinspected or proved at established intervals.

30.4.2 Use of contractor's inspection equipment.- The contractor's gages, measuring and testing devices shall be made available for reasonable use by the Government when required to determine conformance with contract requirements. If conditions warrant, contractor's personnel shall be made available for operation of such devices and for verification of their accuracy and condition.

30.5 Control of subcontracted supplies.-

30.5.1 Responsibility.- The contractor is responsible for assuring that all supplies and services to apply on Government contracts conform to the contract requirements whether manufactured or processed by the contractor or procured from subcontractors. The selection of sources and the nature and extent of control, including both contractor incoming inspection and surveillance, if any, at the subcontractor's plant, shall be based on and adjusted according to the nature of the supplies, the quality evidence furnished by the subcontractor and his demonstrated capability to perform in the specialized field involved. To assure an adequate and economical system for the control of purchased material, the contractor shall utilize to the fullest extent practical, objective evidence of quality furnished by the subcontractor.

30.5.2 Subcontract data.- The contractor shall assure that applicable requirements are properly included or referenced in all subcontracts for supplies ultimately to apply on a Government contract. Subcontracts shall contain at least the following information:

(a) The applicable Government contract number, name, and address of the subcontractor and the consignee.
(b) A clear description of the supplies ordered, including as applicable:
    (1) Specifications, drawings, process requirements, preservation and packaging requirements, classification of defects, inspection instructions (including requirements for contractor inspection of subcontractor's facility) and other necessary data.
    (2) Data necessary when provision is made for direct shipment from the subcontractor to Government activities.

30.5.3 Government inspection of subcontracts.- The Government reserves the right to inspect, at source, supplies or services not manufactured or performed within the contractor's facility. Government subcontract inspection shall not constitute acceptance, nor shall it relieve the contractor of his responsibility to furnish an acceptable end item. The purpose of this inspection is to assist the Government representative at the contractor's facility to determine the conformance of supplies or services with contract requirements. Such inspection can only be requested by or under authorization of the Government representative.

30.5.3.1 When Government subcontract inspection at source is required, the contractor shall add to his subcontract the following statement:

> "Government inspection is required prior to shipment from your plant. When material is ready for inspection or, if practical, ten days in advance thereof, notify the Government representative who normally services your plant".

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30.5.3.2 When under authorization of the Government representative, copies of the subcontract are to be furnished directly by the subcontractor to the Government representative at his facility rather than through Government channels, the contractor shall add to his subcontract substantially as follows:

> "On receipt of this order, promptly furnish a copy to the Government representative who normally services your plant, or, if none to the nearest Army, Navy or Air Force inspection office in your locality. In the event the representative or office cannot be located, our purchasing agent shall be notified immediately".

30.5.3.3 Review and processing of subcontracts.- All subcontracts and referenced data for supplies applying to a Government contract shall be available for review by the Government representative to determine compliance with the requirements for the control of such purchases. Copies of subcontracts required for the Government purposes shall be furnished in accordance with the instructions of the Government representative.

30.5.3.4 Receiving inspection.- Subcontracted supplies shall be subjected to inspection after receipt, as necessary, to assure conformance to contract requirements. In adjusting such inspection consideration shall be given to the control exercised by the subcontractor at source and evidence of sustained quality conformance. The contractor shall provide procedures for withholding from use all incoming supplies pending completion of required tests or receipt of necessary test reports, except that supplies may be released when under positive control. The contractor shall initiate corrective action with his subcontractors upon receipt of nonconforming supplies, whether or not Government source inspected, as indicated by the nature and frequency of the nonconformance. The contractor shall report to the Government representative any nonconformance found on Government source inspected supplies and shall require the subcontractor to coordinate with his Government representative on corrective action.

30.6 Inspection during manufacture.- The contractor shall establish and maintain inspection at appropriately located points in the manufacturing process to assure continuous control of quality of parts, components and assemblies. Inspection check lists shall be provided for personnel to use.

30.7 Special processes.- When Government approved or certification of processes, equipment or personnel is required under the contract, the contractor shall assure that he and his subcontractors are fully qualified prior to requesting Government approval.

30.8 Inspection of completed supplies.- The contractor shall inspect completed supplies as necessary to assure that contract requirements have been met. Inspection check lists shall be provided for personnel to use.

30.9 Sampling inspection.- Any sampling procedures, in addition to those required by contract, used by the contractor to determine the acceptability of supplies, shall afford reliable assurance of the maintenance of acceptable quality levels.

30.10 Indication of inspection status.- The contractor shall maintain a system for identifying the inspection status of supplies. Identification may be accomplished by means of stamps, tags, routing cards, move tickets, tote box cards or other normal control devices. Such controls shall be of a design distinctly different from Government inspection identification.

30.11 Nonconforming supplies.- Procedures shall be provided for control of nonconforming supplies, including procedures for the identification, presentation, and disposition of reworked, repaired or waived supplies. The acceptance of nonconforming supplies is a prerogative of and shall be as prescribed by the Government. All nonconforming supplies shall, when practicable, be diverted from normal material movement channels. The nonconforming supplies shall be positively identified to prevent use until disposition is made. Holding areas mutually agreeable to the contractor and Government representative shall be provided.

30.12 Storage.- The contractor shall provide adequate procedures for control of supplies stored for the Government or to be applied to Government contracts to insure preservation and treatment in accordance with applicable requirements. Procedures shall define inspections to be conducted at scheduled intervals.

30.13 Quality control records.- The contractor shall maintain adequate records throughout all stages of contract performance of examination and tests, including checks made to assure accuracy of examination and testing equipment and other control media. All quality control records shall be available for review by the Government representative and copies of individual records shall be furnished him upon request.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 68 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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